

RAGHAVENDRA DINESH

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<https://github.com/rdinesh207> | <https://rdinesh207.github.io/portfolio>

Summary

Applied AI Engineer with over 4 years of experience delivering production-grade agentic LLM systems, RAG pipelines, and multimodal vision-language solutions. Skilled at integrating generative AI into cloud-native applications using LangChain, FastMCP, and Azure AI Foundry, and optimizing inference with structured prompting and low-latency architectures on GCP and Azure. Proven track record of reducing query latency and deployment costs while enhancing user engagement, seeking to drive AI-native product innovation.

TECHNICAL SKILLS

- **Languages & APIs:** Python, C++, FastAPI, REST APIs, SQL
- **Applied AI / ML:** LLM Applications, Agentic Systems, RAG, Computer Vision, Vision-Language Models, Deep Learning, Model Distillation, Transfer Learning, Fine-tuning, Few-shot Learning, Zero-shot Learning
- **Frameworks & Tools:** PyTorch, LangChain, FastMCP, OpenCV, TensorFlow, ONNX, Pinecone
- **Cloud & DevOps:** Docker, GitHub Actions (CI/CD), Azure AI Foundry, GCP Cloud Run
- **Data & Platforms:** Supabase, Azure AI Search

PROFESSIONAL EXPERIENCE

Indra Solutions LLC | *Applied AI Engineer*

May 2025 - Present

- Designed, built, and deployed agentic LLM systems for social media automation using FastMCP and LangChain, running on GCP Cloud Run with Docker, reducing manual marketing effort and improving audience engagement.
- Built production-grade retrieval-augmented generation pipelines (RAG) using Azure AI Foundry, Azure AI Search, LangChain, and Supabase, reducing end-to-end query latency from ~180 seconds to ~12 seconds.
- Developed internal platforms and APIs to support organization-wide collaboration and knowledge access, improving developer productivity and accessibility.
- Optimized CI/CD pipelines using GitHub Actions and container image optimization, reducing Azure container costs.
- Mentored engineers on LLM apps, Supabase optimization, and agentic workflows, improving speed and code quality.

Logos Robotics Lab | *Applied ML Engineer*

Jun 2024 - May 2025

- Built and integrated multimodal (vision-language) AI systems for plant health diagnostics, leveraging zero-shot learning with BLIP-2, SWIN Transformer, and PyTorch-based pipelines to achieve up to 98% classification accuracy.
- Designed a vision-language reasoning pipeline using Q-Former to connect a frozen vision encoder with LLaMA 3.2 (3B), generating localized treatment recommendations with a RougeL F1 score of 0.86.
- Optimized and fine-tuned large models using PEFT, improving inference while reducing computational overhead.
- Validated system across multiple model configurations and datasets, focusing on reliability and practical deployment.

Bosch Global Software Technologies Pvt. Ltd. | *Automation Engineer*

Sep 2019 - Aug 2022

- Applied model distillation to refine an Autoencoder anomaly detection system, reducing analysis time from 30-50 hours to 6-8 hours and enhancing work throughput.
- Designed a smart inventory tracking system using transfer learning on YOLOv4-tiny and Tesseract OCR, enabling real-time tracking and reducing asset search labour by 8 hours per week.
- Accelerated report generation, reducing time by 90% by building a web app to automate data extraction and rendering.

PROJECTS

AI Facebook Shop Agent (MCP + GPT-4o) | github.com/rdinesh207/facebook-shop-agent

2026

- Built a full-stack AI commerce assistant using FastAPI, LangGraph (ReAct), MCP, Supabase, and OpenAI GPT-4o, enabling natural language management of Facebook Shop products, orders, and sales via **Meta Business SDK**.
- Designed a scalable MCP tool layer (13 tools) with real-time SSE streaming and secure OAuth/JWT flow, reducing manual dashboard effort and enabling fast, cache-optimized, and auditable shop operations.

Lung Image Analysis for Disease Detection | github.com/rdinesh207/Lung-disease-segmentation

2023

- Built a multi-task computer vision system for chest X-ray classification, localization, and segmentation using Swin Transformer, Faster R-CNN, UPerNet, and U-Net.
- Achieved strong performance across AUC, IoU, and FROC metrics on VinBigData CXR, SIIM-ACR Pneumothorax, MedMNISTv2, and MINIJSRT datasets.

EDUCATION

Arizona State University

Masters, Artificial Intelligence-Robotics and Autonomous Systems (GPA:3.97)

Aug 2022 - May 2024

Tempe, AZ, USA

PES Institute of Technology

Bachelor of Engineering, Electronics and Communication Engineering (GPA: 3.2)

Aug 2015 - Jul 2019

Bengaluru, India